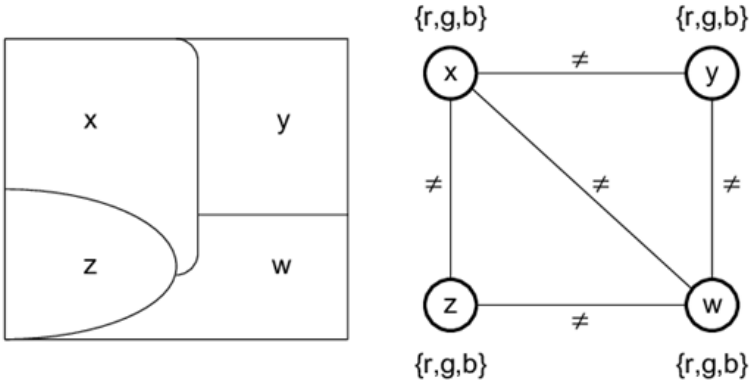
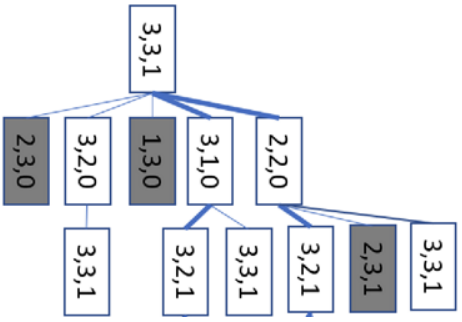
**Sample Questions on AI**

1.. Encode the following graph colouring problem into a constraint satisfaction problem (CSP) having a finite set of variables, a finite domain for each variable, and a set of constraints. In this problem each vertex of the following graph has to be coloured by either red, green, or blue, such that no two adjacent vertices share the same colour. Solve the problem using backtracking considering the order in which you choose variables and their values, which may have a huge impact on search space size. Apply arc-consistency for earlier detection of failure.



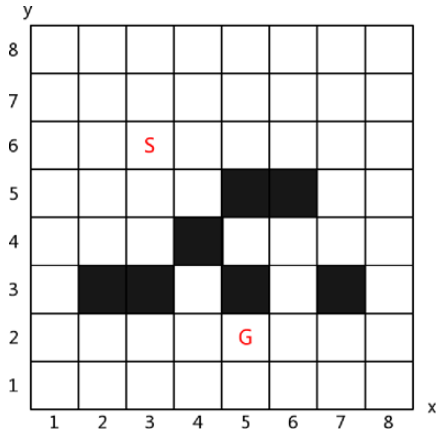

2. What are the advantages of iterative deepening search over breadth-first search? Complete the explicit problem description (shown up to level 2 in the figure below) of the "Missionaries and Cannibals" problem (with 3 members from each group), where count of Missionaries and Cannibals on the left bank is only taken to the consideration. Explain reachability, and solvability with respect to this problem



3. What is consistent heuristic? Consider the following problem of path planning on grids, where a robot has to find a path from the S (start) position to the G (goal) position traversing along the clear grids (i.e., without traversing the dark blocked grids) only. The robot is allowed to move only right, left, up, and down, but not diagonal. The robot is also allowed to move to a clear grid as many times as required, but the less the better in respect of total path cost. (5+13) [CLO3]

i. Step-wise perform A* (state space) search to find the goal position. Use the Manhattan distance as a heuristic to guide the search.

ii. Determine the total path cost (considering the cost of each move is one), total nodes expanded, and total nodes visited by the A* search.




4. What shortcoming of the A* algorithm does the Recursive Best-First Search (RBFS) algorithm address? In the following search tree, the numbers by the nodes denote the f-cost (sum of path cost from the root node to the respective node, and heuristic). The boxed nodes are goals. Describe in detail the way in which the RBFS algorithm searches this tree. Your answer should indicate the order in which nodes are expanded, the reason that this order is used, and should state which of the three goals is found and why. Note that smaller numbers represent more desirable nodes..

